

# RAILWAY TRAFFIC ANALYSIS

Sophie Pull  
Dixant Pant  
Anthony Oparaocha  
George Hunt  
Minh Le  
Mohsin Zeeshan Syed



26TH FEBRUARY 2025  
BETA PRESENTATION

# Project Brief

---

- Modelling of railway data in order to gain a deeper understanding of the variables that affect the probability of railway incidents.
  - Incidents include: derailments, collisions, obstructions etc.
  - Variables include: weather conditions, temperature, track type, train type etc.
- Allows our **client - Synoptix** (an engineering consultancy) - to develop technologies to mitigate these variables and justify the value of their technologies to their customers.

# Synoptix

As **Synoptix**, we want to identify variables that increase railway incident probability, so that we can **develop targeted, data-driven technologies**, as well as **demonstrate and justify their value and effectiveness to stakeholders**.



# Railway Companies

As a **railway company**, we want to **reduce incidents** and associated **costs**, such as rail repairs, train repairs, and insurance expenses, so that we can **prioritise safety**, **prevent casualties**, **build public trust**, and maintain a reliable, **positive brand image**.

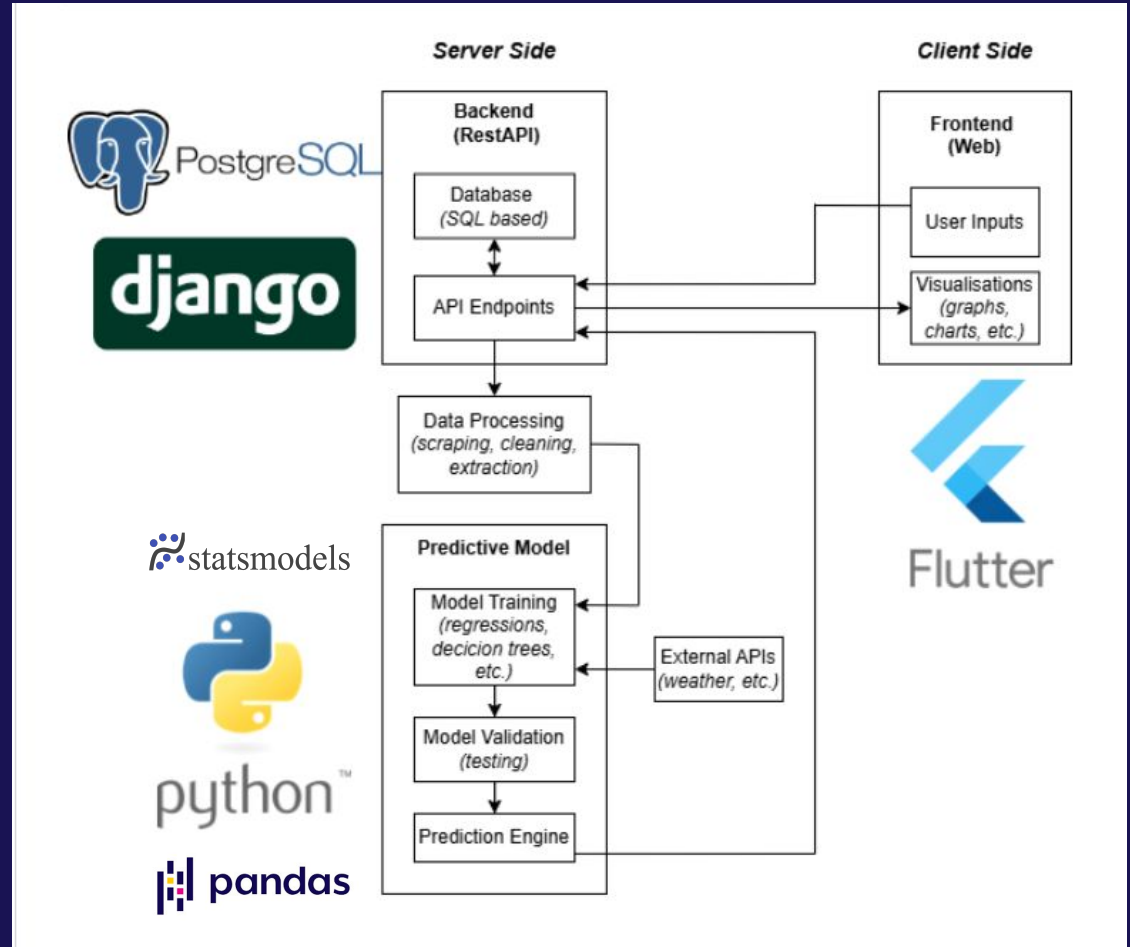


# Railway Commuters

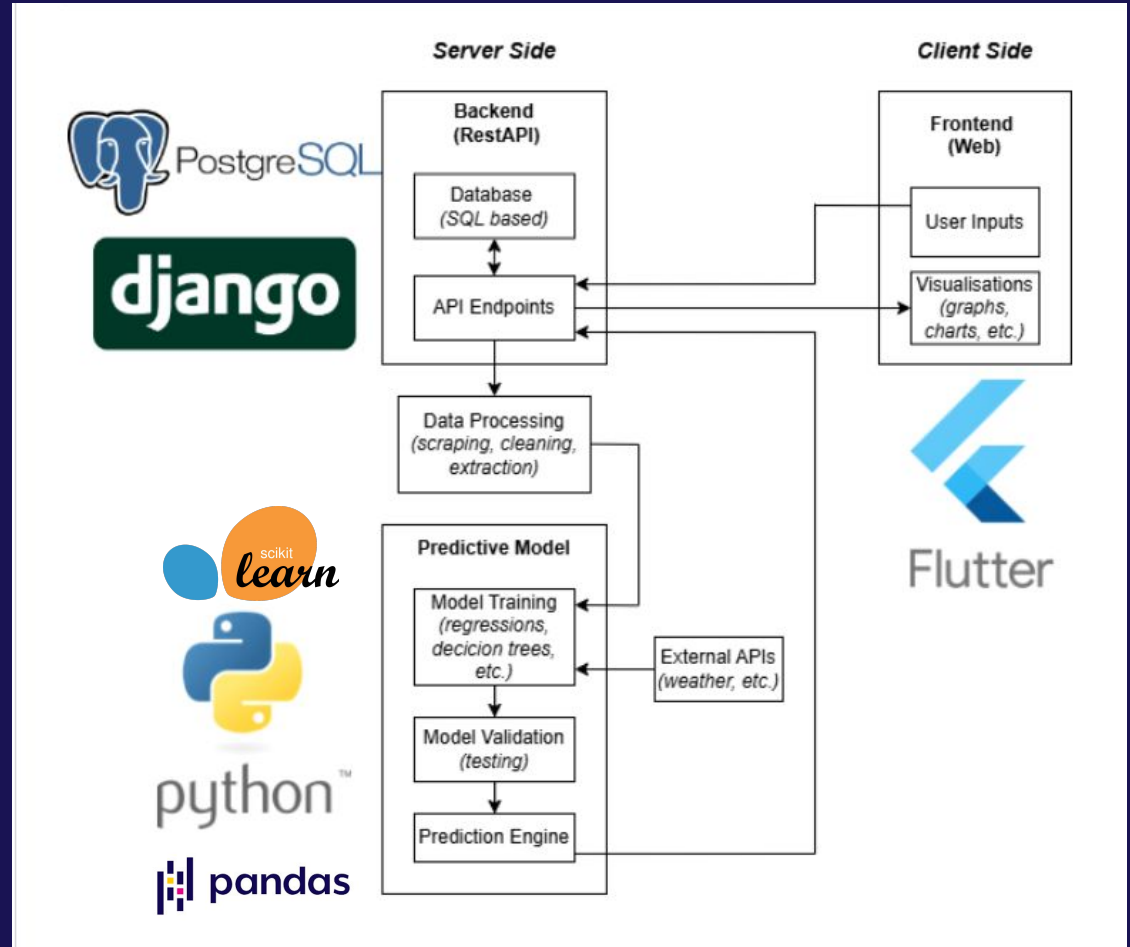
As a **railway commuter**, I want companies to adopt Synoptix technology to prevent incidents, so that I can feel **safer and more confident** when using the railway system.



# Tech Stack + Architecture - *Old*



# Tech Stack + Architecture - *New!*



# CI/CD

---



- Introduced continuous deployment
  - Hosted on an EC2 instance
  - Dockerized versions of Flutter, Django and PostgreSQL
  - Includes a CD pipeline that updates in main in the EC2 instance and updates Docker containers
- The project is publicly accessible at <https://railwaytrafficanalysis.co.uk>
- Used Cloudflare to provide SSL certification to ensure secure connections
- Comprehensive tests in continuous integration
  - Code coverage

# Backend Code Coverage:

# 84%

Coverage report: 84%

Files Functions Classes

coverage.py v7.6.12, created at 2025-02-25 10:07 +0000

| File                                                                      | statements | missing    | excluded | coverage   |
|---------------------------------------------------------------------------|------------|------------|----------|------------|
| manage.py                                                                 | 11         | 2          | 0        | 82%        |
| rtapi/__init__.py                                                         | 1          | 0          | 0        | 100%       |
| rtapi/admin.py                                                            | 12         | 0          | 0        | 100%       |
| rtapi/apps.py                                                             | 4          | 0          | 0        | 100%       |
| rtapi/management/__init__.py                                              | 0          | 0          | 0        | 100%       |
| rtapi/management/commands/__init__.py                                     | 0          | 0          | 0        | 100%       |
| rtapi/management/commands/clean_data.py                                   | 39         | 5          | 0        | 87%        |
| rtapi/migrations/__init__.py                                              | 0          | 0          | 0        | 100%       |
| rtapi/migrations/0001_initial.py                                          | 6          | 0          | 0        | 100%       |
| rtapi/migrations/0002_userinput_mainaccidentcauses_description.py         | 4          | 0          | 0        | 100%       |
| rtapi/migrations/0003_remove_mainaccidentcauses_description.py            | 4          | 0          | 0        | 100%       |
| rtapi/migrations/0004_mainaccidentcauses_description_and_more.py          | 4          | 0          | 0        | 100%       |
| rtapi/migrations/0005_userinput_gross_tonnage.py                          | 4          | 0          | 0        | 100%       |
| rtapi/migrations/0006_rename_track_direction_userinput_train_direction.py | 4          | 0          | 0        | 100%       |
| rtapi/models.py                                                           | 134        | 11         | 0        | 92%        |
| rtapi/multinomial_logit.py                                                | 32         | 25         | 0        | 22%        |
| rtapi/random_forest_model.py                                              | 37         | 13         | 0        | 65%        |
| rtapi/serializers.py                                                      | 46         | 0          | 0        | 100%       |
| rtapi/tests/__init__.py                                                   | 0          | 0          | 0        | 100%       |
| rtapi/tests/test_scripts.py                                               | 120        | 2          | 0        | 98%        |
| rtapi/tests/tests.py                                                      | 104        | 0          | 0        | 100%       |
| rtapi/urls.py                                                             | 5          | 0          | 0        | 100%       |
| rtapi/views.py                                                            | 93         | 43         | 0        | 54%        |
| rtapi/__init__.py                                                         | 1          | 0          | 0        | 100%       |
| rtapi/asgi.py                                                             | 4          | 4          | 0        | 0%         |
| rtapi/settings.py                                                         | 29         | 1          | 0        | 97%        |
| rtapi/urls.py                                                             | 6          | 0          | 0        | 100%       |
| rtapi/wsgi.py                                                             | 4          | 4          | 0        | 0%         |
| <b>Total</b>                                                              | <b>708</b> | <b>110</b> | <b>0</b> | <b>84%</b> |

coverage.py v7.6.12, created at 2025-02-25 10:07 +0000

# Frontend Code Coverage:

# 75%

## LCOV - code coverage report

Current view: [top level](#)

Test: [lcov.info](#)

Test Date: 2025-02-25 10:29:10

|            | Coverage | Total | Hit |
|------------|----------|-------|-----|
| Lines:     | 75.2 %   | 725   | 545 |
| Functions: | -        | 0     | 0   |

| Directory                   | Line Coverage ↕               |       |     |
|-----------------------------|-------------------------------|-------|-----|
|                             | Rate                          | Total | Hit |
| <a href="#">lib/</a>        | <div><div></div></div> 73.1 % | 644   | 471 |
| <a href="#">lib/pages/</a>  | <div><div></div></div> 92.5 % | 80    | 74  |
| <a href="#">lib/themes/</a> | <div><div></div></div> 0.0 %  | 1     |     |

Note: 'Function Coverage' columns elided as function owner is not identified.

Generated by: [LCOV version 2.3-1](#)

# Interface Designs (Old)

## Railway Traffic Analysis

Please enter risk factor parameters into the fields below. We will then determine the associated risks based on these factors.

Amount of rain (mm):

Temperature (degrees C):

Proximity of crossing (km):

Train speed (mph):

Wind speed (mph):

SYNOPTIX

SYNOPTIX

Railway Traffic Analysis

### Train External Data

Type of weather

Clear

Precipitation

mm

Temperature

°C

Location

Time

Time

### Train Internal Data

Type of Train

Train Speed 0-250 Mph

Accident in the last 5 years ☐

Angle of the Train

Number of Signals

- No protection about data being entered by the user.
- No dropdown options to help user know what to input
- Not very aesthetically pleasing

# Interface Designs (Old)

SYNOPTIX

Railway Traffic Analysis

Train External Data

Weather Condition

Cloudy

Track Type

Visibility

Track Direction

Signalization

Train Internal Data

Equipment

Method of Operation

Submit

MVP Interface

SYNOPTIX

Railway Traffic Analysis

Train External Data

Weather Condition

Track Type

Visibility

Track Direction

Signalization

☒ °F

☐ °C

Temperature

0/3

Train Internal Data

Equipment

Method of Operation

☒ mph

☐ km/h

Train Speed

0/3

Equipment attended

Submit

Testing Day

# Testing Day feedback

---

What went well:

- Intuitive user interface
- Users like the dropdown menus and input validation (e.g no one was able to put text into numerical fields!)

Issues which we have since addressed (see Product Demo shortly!)

- A few user input bugs which we fixed for Beta (temperature input value validation, allow decimals, allow user to change fields back to being empty)
- Students unaware of train definitions (not a problem for Synoptix but we fixed this anyway)
- Unit buttons (change between F and C) in unintuitive position
- No user instructions

# PROJECT DEMO

[railwaytrafficanalysis.co.uk](http://railwaytrafficanalysis.co.uk)

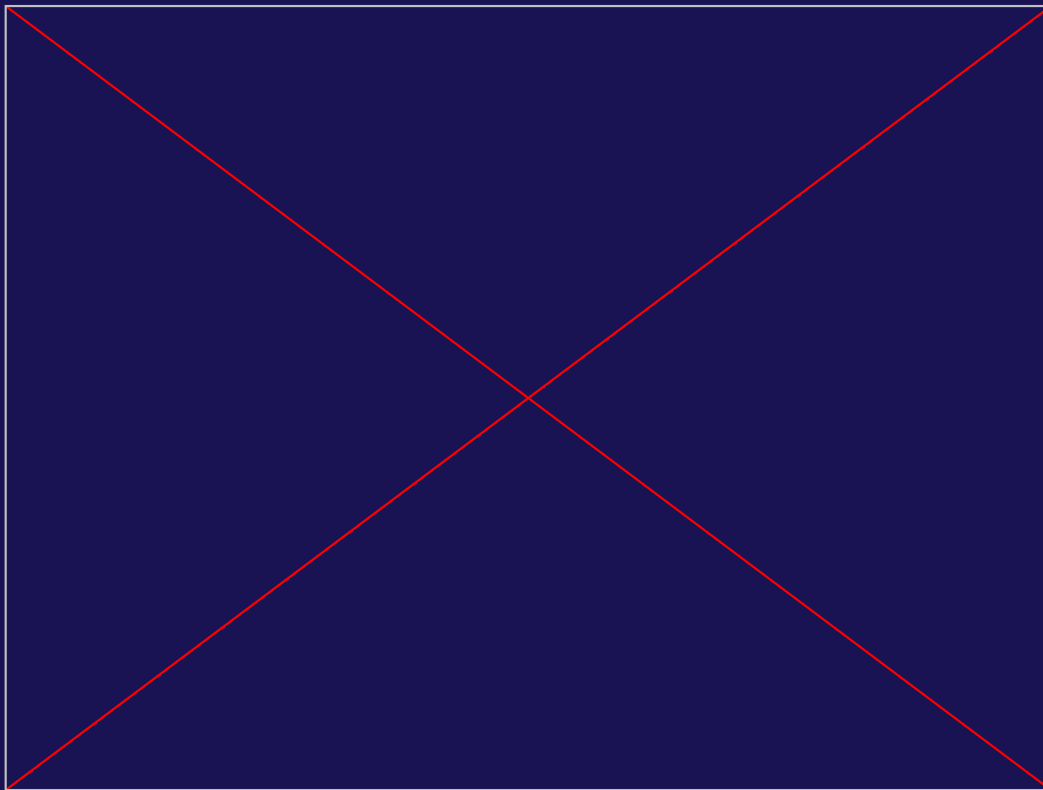
# Testathon feedback + Current Project Demo

---

Implemented changes  
from **Testing day**  
feedback

Tested whether these  
changes were  
effective at the  
**Testathon**

Issues to address:  
Using different colours  
on the output page



# Current Project Demo (cont.)

localhost:53424

SYNOPSIS

Railway Traffic Analysis

Information

User Instructions

Light and Dark Mode

Train External Data

Weather condition

Track type

Visibility

Train direction

Signalization

Method of Operation

Train Internal Data

Equipment

Train speed

Equipment attended

SUBMIT

DEV

# Predictive Model - Old



| MNLogit Regression Results                 |                                      |                   |           |        |       |         |        |
|--------------------------------------------|--------------------------------------|-------------------|-----------|--------|-------|---------|--------|
| =====                                      |                                      |                   |           |        |       |         |        |
| Dep. Variable:                             | y                                    | No. Observations: | 100       |        |       |         |        |
| Model:                                     | MNLogit                              | Df Residuals:     | -15       |        |       |         |        |
| Method:                                    | MLE                                  | Df Model:         | 110       |        |       |         |        |
| Date:                                      | Tue, 12 Nov 2024                     | Pseudo R-squ.:    | 0.5265    |        |       |         |        |
| Time:                                      | 19:49:54                             | Log-likelihood:   | -79.586   |        |       |         |        |
| converged:                                 | False                                | LL-Null:          | -168.09   |        |       |         |        |
| Covariance Type:                           | nonrobust                            | LLR p-value:      | 5.347e-05 |        |       |         |        |
| =====                                      |                                      |                   |           |        |       |         |        |
|                                            | y=accident_type_Explosive-detonation | coef              | std err   | z      | P> z  | [0.025  | 0.975] |
| -----                                      |                                      |                   |           |        |       |         |        |
| temperature                                |                                      | 0.3426            | 0.173     | 1.978  | 0.048 | 0.003   | 0.682  |
| track_class                                |                                      | -1.2701           | 0.728     | -1.745 | 0.081 | -2.696  | 0.156  |
| track_density                              |                                      | 0.0353            | 0.026     | 1.333  | 0.183 | -0.017  | 0.087  |
| equipment_type                             |                                      | -1.4062           | 0.745     | -1.887 | 0.059 | -2.867  | 0.054  |
| equipment_attended                         |                                      | -13.1520          | 10.938    | -1.202 | 0.229 | -34.590 | 8.286  |
| train_speed                                |                                      | 0.0203            | 0.052     | 0.393  | 0.694 | -0.081  | 0.122  |
| remote_control_locomotive                  |                                      | -2.4196           | 1.934     | -1.251 | 0.211 | -6.209  | 1.370  |
| positive_alcohol_tests                     |                                      | 0.7272            | 0.456     | 1.596  | 0.110 | -0.166  | 1.620  |
| positive_drug_tests                        |                                      | -0.0542           | 0.231     | -0.234 | 0.815 | -0.508  | 0.399  |
| visibility_dawn                            |                                      | 0.3890            | 2.408     | 0.162  | 0.872 | -4.332  | 5.110  |
| visibility_day                             |                                      | -12.8727          | 7.824     | -1.645 | 0.100 | -28.208 | 2.462  |
| visibility_dusk                            |                                      | -7.1932           | 4.673     | -1.539 | 0.124 | -16.353 | 1.966  |
| weather_condition_cloudy                   |                                      | -3.4319           | 4.284     | -0.801 | 0.423 | -11.828 | 4.964  |
| weather_condition_fog                      |                                      | 4.5593            | 3.617     | 1.260  | 0.208 | -2.531  | 11.649 |
| weather_condition_rain                     |                                      | -8.4325           | 8.459     | -0.997 | 0.319 | -25.012 | 8.147  |
| weather_condition_sleet                    |                                      | 4.4000            | 3.872     | 1.136  | 0.256 | -3.190  | 11.990 |
| weather_condition_snow                     |                                      | -12.4009          | 8.452     | -1.467 | 0.142 | -28.966 | 4.165  |
| track_type_main                            |                                      | -12.7713          | 7.401     | -1.726 | 0.084 | -27.278 | 1.735  |
| track_type_siding                          |                                      | -7.3775           | 4.280     | -1.724 | 0.085 | -15.765 | 1.010  |
| track_type_yard                            |                                      | -21.1195          | 10.990    | -1.922 | 0.055 | -42.660 | 0.421  |
| method_of_operation_Direct Train Control   |                                      | 9.7192            | 6.285     | 1.546  | 0.122 | -2.599  | 22.038 |
| method_of_operation_Signal Indication      |                                      | -2.9255           | 3.127     | -0.936 | 0.350 | -9.055  | 3.203  |
| method_of_operation_Yard/Restricted Limits |                                      | 17.2128           | 9.505     | 1.811  | 0.070 | -1.416  | 35.842 |
| -----                                      |                                      |                   |           |        |       |         |        |
|                                            | y=accident_type_Fire/violent rupture | coef              | std err   | z      | P> z  | [0.025  | 0.975] |
| -----                                      |                                      |                   |           |        |       |         |        |
| temperature                                |                                      | 0.0207            | 0.020     | 1.030  | 0.303 | -0.019  | 0.060  |
| track_class                                |                                      | 0.0461            | 0.184     | 0.251  | 0.802 | -0.406  | 0.514  |

```
← → ↺ ⓘ http://127.0.0.1:8000/view_model_result/
Pretty-print ☒

{
  "Derailment": {
    "0": 0
  },
  "Fire/violent rupture": {
    "0": 0.176298708
  },
  "Head on collision": {
    "0": 0.0245289262
  },
  "Hwy-rail crossing": {
    "0": 0.1168114646
  },
  "Obstruction": {
    "0": 0.0366203834
  },
  "Other (describe in narrative)": {
    "0": 0.0971297385
  },
  "Other impacts": {
    "0": 0.377528056
  },
  "RR grade crossing": {
    "0": 0.0240681937
  },
  "Raking collision": {
    "0": 0.0369422781
  },
  "Rear end collision": {
    "0": 0.0425723955
  },
  "Side collision": {
    "0": 0.067499856
  }
}
```

# Predictive Model - New



- Scripts to clean, preprocess and analyze the data using Pandas
  - 220,542 entries to 9,164 entries
- Experimenting with machine learning algorithm - *Random Forest classifier*
  - A form of classification - “*which accident type is it?*” given the input parameters
- Trained the model on clean data
- Highest accuracy so far: 71%
- Future Plans: “*given an accident occurs, what type is it?*” rather than “*will an accident occur here?*”

```
{
  "Accident Type_Broken train collision": 0,
  "Accident Type_Derailment": 0.29,
  "Accident Type_Explosion-detonation": 0,
  "Accident Type_Fire/violent rupture": 0.01,
  "Accident Type_Head on collision": 0.01,
  "Accident Type_Hwy-rail crossing": 0.15,
  "Accident Type_Obstruction": 0.24,
  "Accident Type_Other (describe in narrative)": 0.1,
  "Accident Type_Other impacts": 0.14,
  "Accident Type_RR grade crossing": 0,
  "Accident Type_Raking collision": 0.01,
  "Accident Type_Rear end collision": 0.02,
  "Accident Type_Side collision": 0.03
}
```

# Issues We Faced - Technical

---

- **Train Jargon**
  - Train-specific knowledge and terminology
  - Frontend - changed often as we learn new information about train terms
    - e.g 'method of operation' refers to the *method* used to *signalise*
  - Script to clean data - unsure how to identify outliers
    - e.g lots of entries had Strings for 'track density' (a numerical field) - assumed this was erroneous, but actually this is allowed
- Learning how to do ML

# Issues We Faced - Project Management

---

- **Unequal workload in TB1**
  - Some team members felt unsure and uncomfortable contributing due to a lack of knowledge of what was going on in the backend.
  - Team members weren't independently creating and breaking down their own issues, leading to uneven workload distribution.
- **Kanban board not being used effectively**
  - Not used often by some team members, so unsure what everyone is working on until PR is made
  - Not assigning themselves tasks
  - Assigning themselves tasks for the future but not actually being worked on yet, so others could not pick up those tasks

# Current Project Management

---

- Introduced **alternating project managers**
- Introduced **sprints**
- **In-person** weekly group programming instead of virtual
- Doing **proper code reviews** - less afraid to have discussions on PRs
- Better **documentation** - NumPy style and reST docstrings
- Messaging on **Teams** instead of WhatsApp

# Old MVP Future Plans (Backend)

---

## Improve model

- ✓ Sanitize data!
- ✓ Add more predictor variables
- ⌚ Confidence intervals
- ✓ Improve performance

## Experiment with different models

- ⌚ Use different statistical models
- ✓ Experiment with ML

# *Future Plans for product - Backend*

---

## Login System

Using JWT authentication to allow for offline capabilities (per client request)

## Offline capabilities

Be able to retrieve previous queries to model

Per clients request

## Improve model accuracy

Look into feature extraction

Will require researching literature - *may take a while to learn more advanced ML techniques!*

## Explore 'two step' prediction

First step: *will* an accident occur?

Second step: given an accident occurs, what accident type is it?

Will require researching more literature

# Old MVP Future Plans (frontend)

## Enhancing input options:

✓ Add input fields for more predictor variables.

## Model and Data Selection:

Allow users to choose **✗ (not really possible, lack of accessible data)** specific dataset and **⌚ (to be implemented)** models.

## Prediction Comparison:

✓ **(UI mockup)** Enable side-by-side prediction comparison.

## Improving Visualisation:

✓ **(UI mockup)** Use advanced, interactive graphs.

## User Login System:

⌚ **(in progress)** Add a login system to save submission history.

# *Future Plans for product - Frontend*

---

## Login System

Make login pages for the user

## View previous searches

Add pages to be able to view the previous queries to model and the predictions generated

## Interactive Graphs

See next slides for mockups

Should be able to edit the variables and view how this changes the probability, on the same page.

## Improved visualisations

See next slides for mockups

Display various statistics on the data

# Future Plans - UI mockups

Derailement

**18.5%**

⬇️ 3.3% vs historical

Head-on collision

**7.2%**

⬆️ 0.9% vs historical

Side collision

**15.3%**

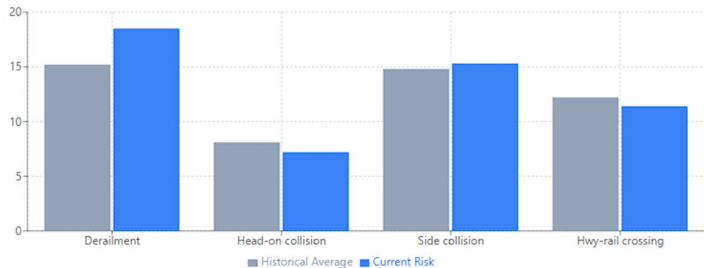
⬇️ 0.5% vs historical

Hwy-rail crossing

**11.4%**

⬆️ 0.8% vs historical

Historical vs Current Risk Comparison



⚠️ Current Risk Level

**18.5%**

15% higher than average for these conditions

⚙️ Variables

Train Speed (km/h)



44 km/h

Weather Condition

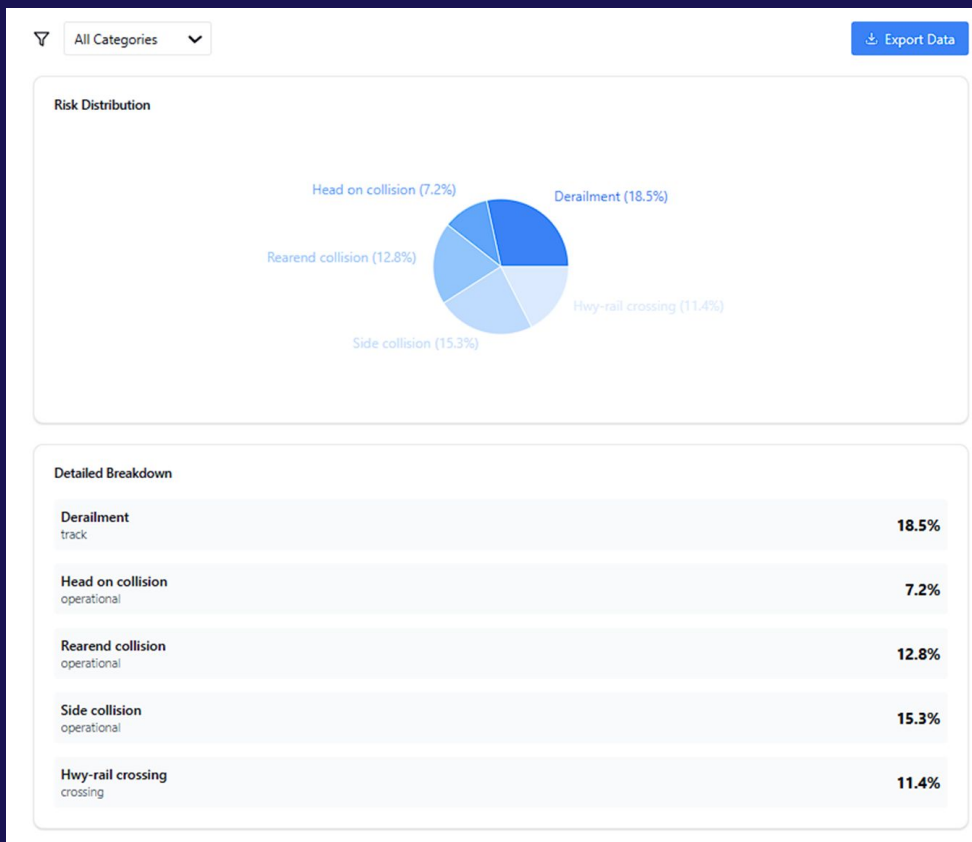
Clear



📌 Recommendations

- Consider reducing speed by 20km/h in current conditions
- Increase track inspection frequency
- Monitor weather conditions for next 4 hours

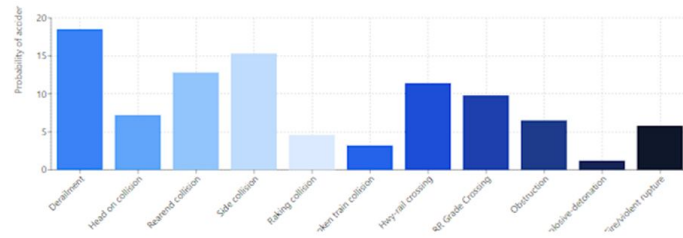
# Future Plans - UI mockups



## Railway Accident Statistics

Real time risk analysis and incident probability distribution

Probability Distribution of Railway Accidents

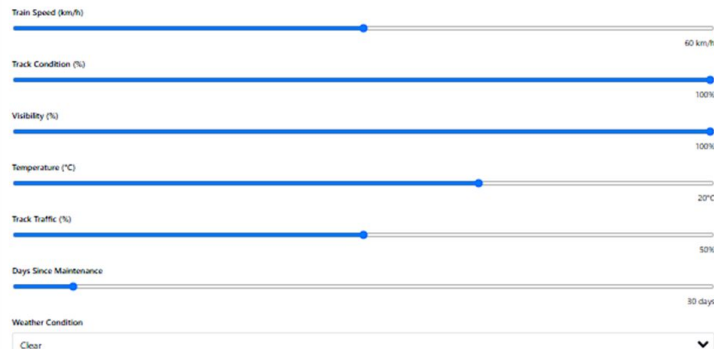


### Current Risk Level

**18.5%**

Highest probability: Derailment

### Risk Variables



[Export Analysis](#)

# Questions